

Programming-I (CS1139)

Lab 1: Introduction to Python & Computational Thinking

Objectives:

- Become familiar with Spyder IDE
- Write, execute and debug simple Python programs
- Practice input, output, arithmetic and conditional logic
- Test programs using valid and invalid inputs

Instructions

- Create a OneDrive folder Lab1 and save all programs meaningfully.
- Write comments on every program.
- Test with at least 3 valid and 2 invalid inputs wherever applicable.
- Predict the output before executing each program.

Lab Problems

1. Write a Python program to display 'Hello JKLU!' and 'I am learning Python.'
2. Accept the student's name and branch as input and display a welcome message.
3. Input the length and width (in metres) of a rectangular plot and calculate its area.
4. Modify Problem 3 to accept dimensions in feet and display the area in square metres.
5. Input two positive integers x and y. Check whether y is divisible by x. For negative inputs, display 'Invalid Input'.
6. Input an integer and determine whether it is even or odd.
7. Input the radius of a circle (1–100 only) and calculate its area. Handle invalid inputs.
8. Input temperature in Celsius and display the equivalent Fahrenheit and Kelvin values.
9. Input the number of seconds (1–86400) and display the equivalent time in HH:MM:SS format.
10. Mini Challenge (Choose ONE):
 - (a) Attendance Percentage Calculator (eligible if attendance $\geq 75\%$), OR
 - (b) Simple Electricity Bill Calculator, OR
 - (c) ATM PIN Validation (4-digit PIN).

Reflection (Submit with Lab)

- What was the easiest program?
- What error did you encounter?
- How did you fix it?
- What is one new thing you learned today?